

Disorders of the Apocrine Sweat Glands

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Apocrine glands are adnexal glands distributed in the axillae, anogenital region, Moll's glands of the eyelid, ceruminous glands of the external auditory meatus, and mammary glands. They may also be found in limited numbers on the face and abdomen. These glands remain quiescent until puberty. Embryologically, apocrine glands develop from the upper bulge of the hair follicle during the late fourth month of gestation, with continued development coinciding with hair follicle formation. Structurally, apocrine glands consist of three parts: the intraepithelial duct, the intradermal duct, and the secretory portion.

Four conditions are discussed in this chapter: **apocrine bromhidrosis**, **apocrine chromhidrosis**, **Fox-Fordyce disease**, and **hidradenitis suppurativa**. It is important to note that Fox-Fordyce disease and hidradenitis suppurativa are not primary disorders of apocrine glands; rather, apocrine glands become secondarily involved in these conditions.

Apocrine Bromhidrosis

Bromhidrosis refers to malodorous body odor, a common occurrence in postpubertal individuals. When the odor becomes excessive or particularly offensive, it is termed **apocrine bromhidrosis**, which originates from apocrine gland secretions—most commonly in the axillae. This condition can significantly impair psychosocial functioning. The terms *bromhidrosis* and *osmidrosis* are often used interchangeably, though some sources define *osmidrosis* as offensive odor and *bromhidrosis* as osmidrosis occurring in conjunction with hyperhidrosis.

Epidemiology

Onset typically occurs after puberty and is more prevalent among African American populations. There is no specific geographic predilection, although symptoms may worsen during summer months or in warm climates. Poor personal hygiene can exacerbate the condition. A familial tendency has been observed, particularly in individuals of Far Eastern descent.

Etiology and Pathogenesis

Apocrine glands are most densely concentrated in the axillae and genital region, with additional presence in the breasts, ears (ceruminous glands), and periorbital areas (Moll's glands). The characteristic odor in bromhidrosis results from bacterial degradation of apocrine secretions.

The axillary microbiome predominantly includes gram-positive bacteria, with aerobic *Corynebacterium* species playing a key role in odor production. These bacteria metabolize apocrine gland secretions, generating volatile compounds such as **16-androstenes**, **5 α -androstanol**, and **5 α -androstenone**, which are known contributors to unpleasant odor.

Bacterial action also produces **ammonia** and **short-chain fatty acids**. The most well-characterized of these is **(S)-3-methyl-2-hexenoic acid**, a major component responsible for the characteristic axillary odor. The precise biotransformation pathways of these steroids and fatty acids remain incompletely understood and require further investigation.